

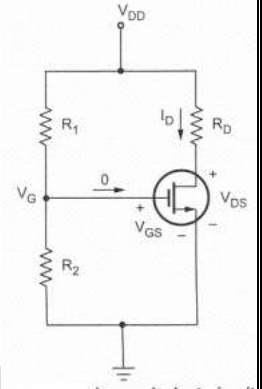
Module 2 (a) - MOSFETs : Biasing in MOS Amplifier Circuits

Syllabus - Biasing in MOS amplifier circuits: Fixing V_{GS} , Fixing V_G , Drain to Gate feedback resistor.

Introduction : Biasing is needed to establish an appropriate dc operating point for the MOS transistor. Biasing provides a stable and predictable dc drain current I_D by dc drain-to-source voltage V_{DS} which ensures the operation of the MOSFET in the **Saturation region** for all expected input-signal levels.

2.1 - Biasing by Fixing V_{GS}

The simplest approach for biasing a MOSFET is to fix its gate-to-source voltage V_{GS} to the value required to provide the desired I_D . This voltage value can be derived from the power-supply voltage V_{DD} through the use of an appropriate voltage divider network as shown in figure. Since $I_G = 0$; $V_G = V_{GS} = \frac{V_{DD} R_2}{R_1 + R_2}$



But this is not a good approach to biasing a MOSFET and the reason for the same are given below.

(a) The drain current I_D can be expressed as

$$I_D = \left[\frac{1}{2} \mu_n C_{ox} \right] \left[\frac{W}{L} \right] (V_{GS} - V_t)^2$$

The threshold voltage V_t , the oxide-capacitance C_{ox} , and the MOSFET aspect ratio $\left(\frac{W}{L}\right)$ are device dependent vary widely among devices of the same size and type. This results change in I_D if the device gets replaced.

(b) Also, the threshold voltage V_t and the carrier mobility μ_n depend on temperature. Hence if the value of V_{GS} is fixed, the drain current I_D becomes very much temperature dependent.

2.2 - Biasing by Fixing V_G and Connecting a Resistance in the Source

An excellent biasing technique for discrete MOSFET circuits consists of fixing the dc voltage at the gate, V_G , and connecting a resistance in the source lead, as shown in Figure.

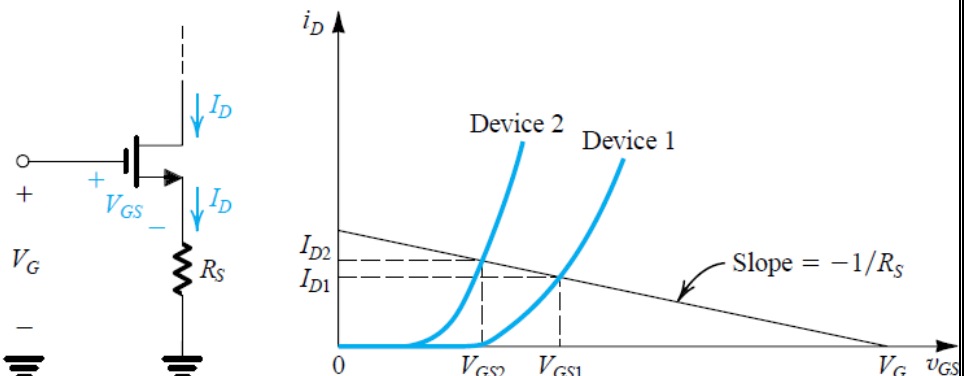
For the circuit shown; $V_G = V_{GS} + I_D R_S$

If $V_G \gg V_{GS}$, then I_D will be mostly determined by the values of V_G and R_S . However, even if V_G is not much larger than V_{GS} , resistor R_S provides negative feedback, which acts to stabilize the value of the bias current I_D .

Consider due to some reason, I_D increases, since V_G is constant, V_{GS} will have to decrease. This in turn results in a decrease in I_D , and hence I_D level is held constant. Thus the action of R_S works to keep I_D as constant as possible.

This negative feedback action of

R_S gives it the name **degeneration resistance**.



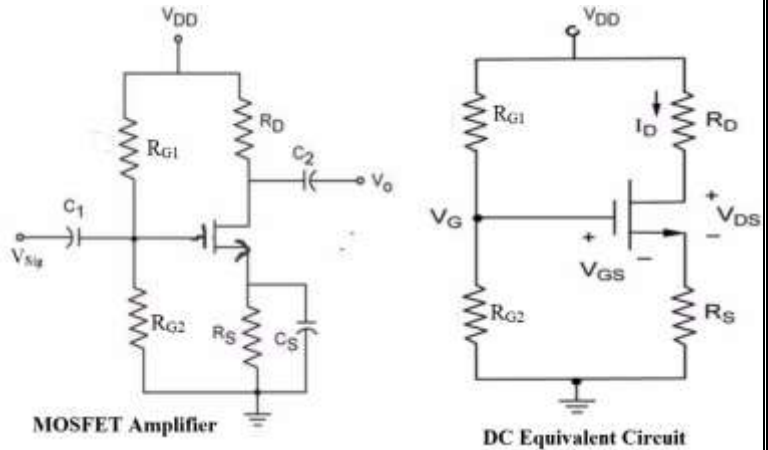
The following are the methods used to bias MOSFET by fixing V_G and Connecting source resistance R_S

2.2(a) - Voltage Divider Bias :

Figure shows a MOSFET amplifier biased by fixing V_G using a single power-supply V_{DD} and which derives V_G through a voltage divider ($R_{G1} - R_{G2}$). Since $I_G = 0$, R_{G1} and R_{G2} can be selected to be very large (in the Meg-ohm range, so that the circuit can give high input impedance).

The resistor R_D is selected to be as large as possible to obtain high gain but small enough to allow for the desired signal swing at the drain while keeping the MOSFET in saturation at all times.

Since $I_G = 0$; $V_G = \frac{V_{DD} R_{G2}}{R_{G1} + R_{G2}}$



Applying KVL to Gate – source loop, $V_G - V_{GS} - I_D R_S = 0$ and $V_S = I_D R_S$

Combining the above equations $R_S = \frac{V_S}{I_D} = \left[\frac{V_G - V_{GS}}{I_D} \right]$

KVL to Drain to source loop; $V_{DD} - I_D R_D - V_D = 0$ and $V_D = V_{DS} + V_S \rightarrow R_D = \left[\frac{V_{DD} - V_D}{I_D} \right]$

The general approximation used is $V_G = V_S = I_D R_D = \left[\frac{V_{DD}}{3} \right]$

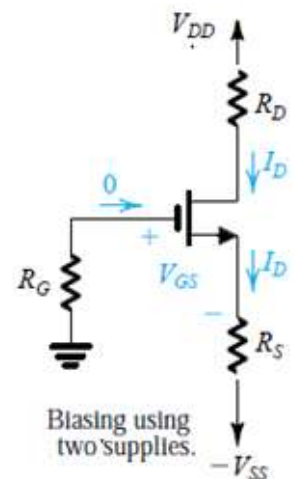
2.2(b) - Using Two Power supplies :

Constant V_G biasing is achieved using two power supplies as shown in the figure.

Resistor R_G establishes a dc ground at the gate and presents a high input resistance to a signal source that may be connected to the gate through a coupling capacitor.

Applying KVL to source – gate loop as well drain to source loop

$-V_{SS} + I_D R_S + V_{GS} = 0$	$V_{DD} - I_D R_D - V_D = 0$
$I_D = \left[\frac{V_{SS} - V_{GS}}{R_S} \right]$	$I_D = \left[\frac{V_{DD} - V_D}{R_D} \right]$



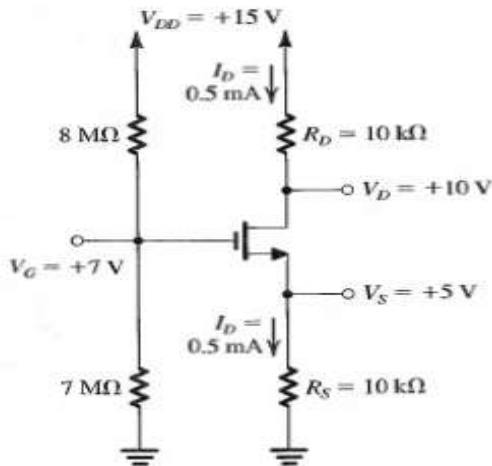
Problem1: It is required to design constant V_G bias circuit using a voltage divider network to establish a dc drain current $I_D = 0.5 \text{ mA}$. The MOSFET is specified to have $V_t = 1 \text{ V}$, $K_n^1(W/L) = 1\text{mA/V}^2$. Neglect the channel-length modulation effect (i.e., assume $\lambda = 0$). Use a power-supply $V_{DD} = 15 \text{ V}$. Calculate the percentage change in the value of I_D obtained when the MOSFET is replaced with another unit having the same $K_n^1(W/L)$, but $V_t = 1.5 \text{ V}$.

Solution: As a thumb rule, the V_{DD} is equally distributed across R_D , R_S and V_{DS} .

Hence $V_S = I_D R_D = V_{DS} = \frac{V_{DD}}{3} = \frac{15}{3} = 5\text{V}$ and $V_D = V_{DS} + V_S = 5 + 5 = 10\text{V}$

$$R_D = \left[\frac{V_{DD} - V_D}{I_D} \right] = \left[\frac{15 - 10}{0.5 \times 10^{-3}} \right] = 10 \text{ k}\Omega$$

$$R_S = \left[\frac{V_S}{I_D} \right] = \left[\frac{5}{0.5 \times 10^{-3}} \right] = 10 \text{ k}\Omega$$



Using the relation, $I_D = \frac{1}{2} K_n' \left(\frac{W}{L} \right) [V_{OV}]^2$

$$V_{OV} = \sqrt{\frac{2I_D}{K_n' \left(\frac{W}{L} \right)}} = \sqrt{\frac{2 \times 0.5 \times 10^{-3}}{1 \times 10^{-3}}} = 1 \text{ V}$$

Also $V_{GS} = V_t + V_{OV} = 1 + 1 = 2 \text{ V}$

and $V_G = V_{RS} + V_{GS} = 5 + 2 = 7 \text{ V}$

$$V_G = \frac{V_{DD} R_{G2}}{R_{G1} + R_{G2}} \quad \text{or} \quad 7 = \frac{15 R_{G2}}{R_{G1} + R_{G2}} \quad \text{or} \quad 7 R_{G1} = 8 R_{G2}$$

Let $R_{G2} = 7 \text{ M}\Omega$; then $R_{G1} = 8 \text{ M}\Omega$

Consider the case where the device is replaced with $V_t = 1.5 \text{ V}$

$$I_D = \frac{1}{2} K_n' \left(\frac{W}{L} \right) [V_{OV}]^2 = \frac{1}{2} K_n' \left(\frac{W}{L} \right) [V_{GS} - V_t]^2 = \frac{1}{2} (1) [V_{GS} - 1.5]^2$$

$$2I_D = [V_{GS} - 1.5]^2 \quad \text{-----(1)}$$

$$V_G = 7 = V_{RS} + V_{GS} = I_D R_S + 2 = 10I_D + V_{GS}$$

$$\text{or} \quad 2I_D = \frac{7 - V_{GS}}{5} \quad \text{-----(2)}$$

Solving equations (1) and (2) results $[V_{GS} - 1.5]^2 = \frac{7 - V_{GS}}{5}$

$$5 [V_{GS}]^2 + 14 V_{GS} - 4.25 = 0 \quad \rightarrow \quad V_{GS} = 2.453 \text{ V} \quad \text{or} \quad V_{GS} = 0.346 \text{ V}$$

Choose $V_{GS} = 2.453 \text{ V}$

$$I_D = \frac{7 - V_{GS}}{10} = \frac{7 - 2.453}{10} = 0.4547 \text{ mA}$$

$$\% \text{ Change in } I_D = \frac{\Delta I_D}{I_D} = \left[\frac{0.4547 - 0.5}{0.5} \right] \times 100 = -9 \%$$

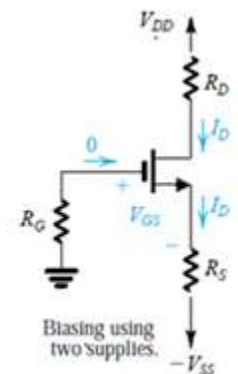
Problem 2: Design the MOSFET bias circuit shown below for a drain current of 0.5 mA and $V_D = +2 \text{ V}$. The supply voltages are $V_{DD} = V_{SS} = 5 \text{ V}$ and the device parameters are $V_t = 1 \text{ V}$, $K_n^1(W/L) = 1 \text{ mA/V}^2$. Neglect the channel-length modulation effect (i.e., assume $\lambda = 0$).

Solution:

$$R_D = \left[\frac{V_{DD} - V_D}{I_D} \right] = \left[\frac{5 - 2}{0.5 \times 10^{-3}} \right] = 6 \text{ k}\Omega$$

$$I_D = 0.5 = \frac{1}{2} K_n' \left(\frac{W}{L} \right) [V_{GS} - V_t]^2 = \frac{1}{2} (1) [V_{GS} - 1]^2 \quad \text{or} \quad V_{GS} = 2 \text{ V}$$

$$R_S = \left[\frac{V_{SS} - V_{GS}}{I_D} \right] = \left[\frac{5 - 2}{0.5 \times 10^{-3}} \right] = 6 \text{ k}\Omega$$



To make $I_G \cong 0$; Choose R_G in the range of $1\text{ M}\Omega$ to $10\text{ M}\Omega$

2.2(c) - Biasing Using Drain to Gate Feedback Resistor

Since $I_G = 0$; a large feedback Resistor R_G is connected between drain and gate will provide the required bias Applying KVL to Drain – to source via gate loop results

$$V_{DD} - I_D R_D - I_G R_G - V_{GS} = 0$$

$$V_{GS} = V_{DD} - I_D R_D \quad \text{or} \quad V_{DD} = V_{GS} + I_D R_D$$

Degeneration Action - If I_D for some reason changes, say increases, then V_{GS} must decrease. The decrease in V_{GS} in turn causes a decrease in I_D , a change that is opposite in direction to the one originally assumed. Thus the negative feedback or degeneration provided by R_G works to keep the value of I_D as constant as possible.

Problem 3: Design the MOSFET bias circuit shown below for a drain current of 0.5 mA . The supply voltages are $V_{DD} = 5\text{ V}$ and the device parameters are $V_t = 1\text{ V}$, $K_n^1(W/L) = 1\text{ mA/V}^2$.

Neglect the channel-length modulation effect (i.e., assume $\lambda = 0$)

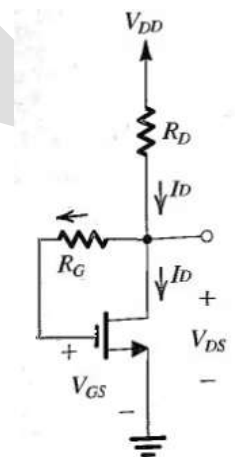
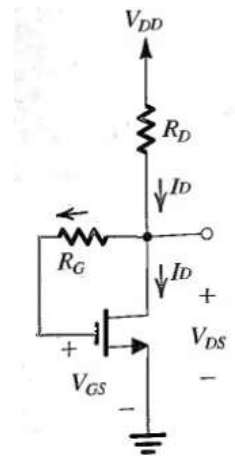
Solution: $I_D = 0.5 = \frac{1}{2} K_n' \left(\frac{W}{L} \right) [V_{GS} - V_t]^2 = \frac{1}{2} (1) [V_{GS} - 1]^2$

$$[V_{GS} - 1]^2 = 1 \quad \text{or} \quad V_{GS} = 1 + 1 = 2\text{ V}$$

Choose $V_{GS} = V_D = 2\text{ V}$

$$R_D = \left[\frac{V_{DD} - V_D}{I_D} \right] = \left[\frac{5 - 2}{0.5 \times 10^{-3}} \right] = 6\text{ k}\Omega$$

To make $I_G \cong 0$; Choose R_G in the range of $1\text{ M}\Omega$ to $10\text{ M}\Omega$



Review Questions

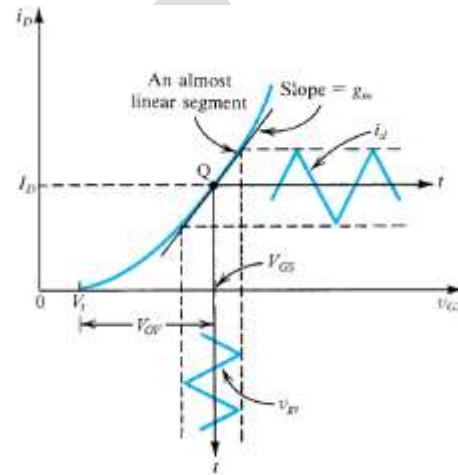
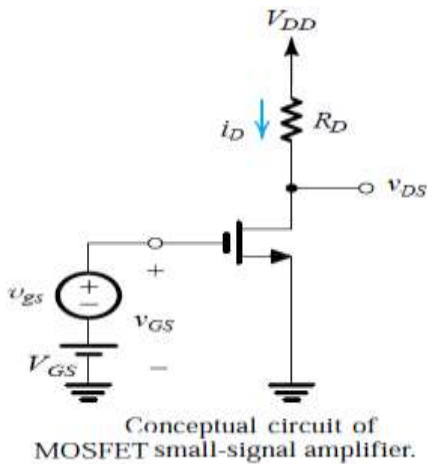
1. Explain the voltage divider bias scheme for biasing MOSFET.
2. State the disadvantage of fixed V_{GS} biasing technique and explain how stability of operating point is achieved in drain to gate feedback resistor biasing technique in a MOSFET amplifier.
3. Explain the MOSFET biasing technique using a large drain-to-gate feedback resistance R_G .

Module 2(b) - Small-Signal Operation and MOSFET Models

Syllabus - Small signal operation and modeling: The DC bias point, signal current in drain, voltage gain, small signal equivalent circuit models, transconductance, The T equivalent circuit model.

2.2(a) - The DC Bias Point:

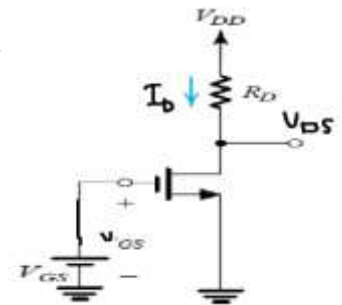
By biasing the MOSFET to operate in the saturation region and by keeping the input signal small, linear amplification can be obtained. Figure shows a conceptual MOSFET amplifier circuit in which MOS transistor is biased by applying a dc voltage V_{GS} , and the input signal to be amplified v_{gs} , is superimposed on the dc bias voltage V_{GS} . The output voltage is taken at the drain.



Neglecting channel length modulation ($\lambda = 0$) and setting $v_{gs} = 0$, the drain current I_D is given as $I_D = \frac{1}{2} k_n [V_{GS} - V_t]^2 = \frac{1}{2} k_n [V_{OV}]^2$. Where $V_{OV} = V_{GS} - V_t$ is the overdrive voltage.

The dc voltage at the drain $V_{DS} = V_{DD} - I_D R_D$

The desired condition is that $[V_{DS} > V_{OV}]$ for MOSFET to be in saturation region for required signal swing.



(a) - The Signal Current in the Drain terminal and transconductance g_m

With the input signal v_{gs} applied. The total instantaneous gate-to source voltage will be

$$V_{GS} = V_{GS} + v_{gs}$$

The total instantaneous drain current i_D is given by

$$i_D = \frac{1}{2} k_n [V_{GS} + v_{gs} - V_t]^2 = \frac{1}{2} k_n [V_{GS} - V_t]^2 + k_n [V_{GS} - V_t] v_{gs} + \frac{1}{2} k_n [v_{gs}]^2 \quad \text{--- (A)}$$

The first term on the right-hand side of Eq. (A) can be recognized as the dc bias current I_D . The second term represents a current component that is directly proportional to the input signal v_{gs} . The third term is a current component that is proportional to the square of the input signal. This last component is undesirable because it represents nonlinear distortion. To reduce the nonlinear distortion introduced by the MOSFET, the input signal should be kept small so that

$$\frac{1}{2} k_n [v_{gs}]^2 \ll k_n [V_{GS} - V_t] v_{gs} \quad \text{or} \quad v_{gs} \ll 2[V_{GS} - V_t] \quad \text{or} \quad v_{gs} \ll 2V_{OV}$$

Under this condition; $i_D \cong \frac{1}{2}k_n[V_{GS}-V_t]^2 + k_n[V_{GS}-V_t] v_{gs} = I_D + i_d$

Where $i_d = k_n[V_{GS}-V_t] v_{gs}$

The transconductance g_m is defined as $g_m = \frac{i_d}{v_{gs}} = k_n[V_{GS}-V_t] = k_n V_{OV}$

The transconductance g_m is equal to the slope of the $i_D - v_{GS}$ characteristic at the bias point and is given as

$$g_m = \left[\frac{\partial i_D}{\partial v_{GS}} \right]_{v_{GS}=V_{GS}}$$

(b) - The Voltage Gain:

The instantaneous drain voltage $v_{DS} = V_{DD} - R_D i_D$

In presence of the signal; $v_{DS} = V_{DD} - R_D (I_D + i_d) = V_{DD} - I_D R_D - i_d R_D = V_{DS} - i_d R_D$

The signal component of the drain voltage $v_{ds} = -i_d R_D = -(g_m v_{gs}) R_D$

The voltage gain $A_V = \frac{v_{ds}}{v_{gs}} = -g_m R_D$

[Note: v_{ds} and v_{gs} are the signal quantities]

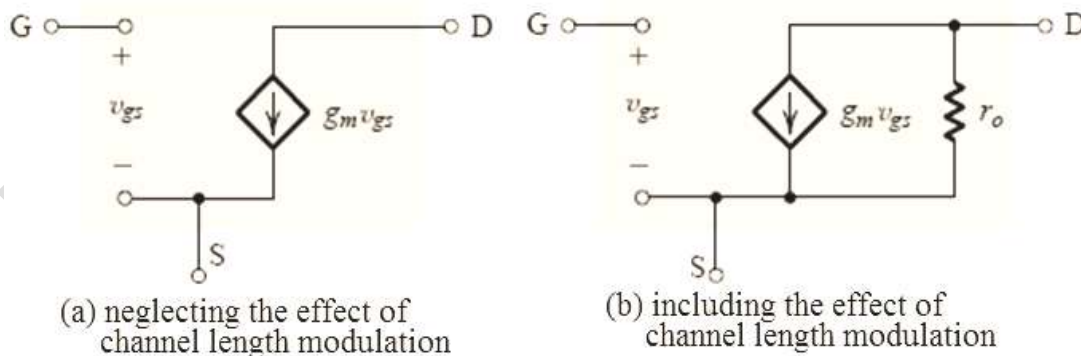
The negative sign indicates that the input and output are phase shifted by 180°

(c) - Small-Signal Equivalent-Circuit MOSFET Models

From a signal point of view, the MOSFET behaves as a voltage-controlled current source. It accepts a signal v_{gs} between gate and source and provides a current $g_m v_{gs}$ at the drain terminal. The input resistance of this controlled source is very high—ideally, infinite. The output resistance—that is, the resistance looking into the drain—also is high (ideally infinite). The small-signal model or a small-signal equivalent circuit of MOSFET is

shown in figure.

Small-signal models for the MOSFET:



In figure (a), the drain current is assumed in saturation to be independent of the drain voltage. However from the MOSFET characteristics in saturation, the drain current does in fact depend on v_{DS} in a linear manner. Such dependence was modeled by a finite resistance r_o between drain and source as shown in figure (b).

The value of r_o is defined as $r_o = \frac{|V_A|}{I_D}$ where $V_A = \frac{l}{\lambda}$

Typically, r_o is in the range of 10 k Ω to 1000 k Ω .

The voltage gain $A_V = \frac{v_{ds}}{v_{gs}} = -g_m (R_D || r_o)$

Thus, the finite output resistance r_o results in a reduction in the magnitude of the voltage gain.

Expressions for Transconductance g_m

Consider the equation $g_m = \frac{i_d}{v_{gs}} = k_n [V_{GS} - V_t]$

Using $k_n = k_n' \left(\frac{W}{L}\right)$

$$g_m = k_n' \left(\frac{W}{L}\right) [V_{GS} - V_t] = k_n' \left(\frac{W}{L}\right) V_{OV} \quad \text{--- (A)}$$

From equation (A), the transconductance g_m is proportional to the (i) process transconductance parameter $k_n' = \mu_n C_{ox}$ (ii) (W/L) ratio of the MOS transistor and (iii) over drive voltage $V_{OV} = V_{GS} - V_t$

Using $V_{OV} = \sqrt{\frac{2I_D}{k_n' \left(\frac{W}{L}\right)}}$ in the transconductance equation,

$$g_m = k_n' \left(\frac{W}{L}\right) \sqrt{\frac{2I_D}{k_n' \left(\frac{W}{L}\right)}} = \sqrt{2I_D k_n' \left(\frac{W}{L}\right)} = \sqrt{2k_n' \left(\frac{W}{L}\right)} \sqrt{I_D} \quad \text{--- (B)}$$

This expression (B) shows two things: (i) For a given MOSFET, g_m is proportional to the square root of the dc bias current. (ii) At a given bias current, g_m is proportional to $\sqrt{\left(\frac{W}{L}\right)}$

Using $k_n' \left(\frac{W}{L}\right) = \frac{2I_D}{[V_{GS} - V_t]^2}$ results

$$g_m = k_n' \left(\frac{W}{L}\right) [V_{GS} - V_t] = \frac{2I_D}{[V_{GS} - V_t]^2} [V_{GS} - V_t] = \left[\frac{2I_D}{V_{GS} - V_t}\right] = \frac{2I_D}{V_{OV}} \quad \text{--- (C)}$$

In summary, there are three different relationships for determining transconductance g_m —Eqns. (A), (B), and (C) and there are three design parameters (W/L) , V_{OV} , and I_D , any two of which can be chosen independently. That is, the designer may choose to operate the MOSFET with a certain overdrive voltage V_{OV} and at a particular current I_D ; the required (W/L) ratio can then be found and the resulting g_m determined.

Problem 1 : A MOSFET is to operate at $I_D = 0.1\text{mA}$ and is to have $g_m = 1\text{mA/V}$. If $k_n' = 50\text{mA/V}^2$. Find the required (W/L) ratio and the overdrive voltage.

Solution : Using $g_m = \sqrt{2k_n' \left(\frac{W}{L}\right)} \sqrt{I_D}$ or $\left(\frac{W}{L}\right) = \frac{g_m^2}{2k_n' I_D} = \frac{1 \times 10^{-6}}{2 \times 50 \times 10^{-3} (0.1 \times 10^{-3})} = \frac{1}{10}$

Also $g_m = \frac{2I_D}{V_{OV}}$ or $V_{OV} = \frac{2I_D}{g_m} = \frac{2(0.1 \times 10^{-3})}{1 \times 10^{-3}} = 0.2\text{ V}$

Problem 2 : Consider the FET amplifier with $V_t = 2\text{ V}$, $k_n' (W/L) = 1\text{ mA/V}^2$, $V_{GS} = 4\text{ V}$, $V_{DD} = 10\text{ V}$, and $R_D = 3.6\text{ k}\Omega$. (a) Find the dc quantities I_D and V_D . (b) Calculate the value of g_m at the bias point. (c) Calculate the value of the voltage gain. (d) If the MOSFET has $\lambda = 0.01\text{ V}^{-1}$, find r_o at the bias point and calculate the voltage gain.

Solution: $I_D = \frac{1}{2} k_n' \left(\frac{W}{L}\right) [V_{GS} - V_t]^2 = \frac{1}{2} (1) [4 - 2]^2 = 2\text{ mA}$

$V_D = V_{DD} - I_D R_D = 10 - 2(3.6) = 2.8\text{ V}$

$g_m = k_n' \left(\frac{W}{L}\right) V_{OV} = 1(2) = 2\text{ mS}$ and $A_v = -g_m R_D = -2(3.6) = -7.2$

$r_o \cong \frac{1}{\lambda I_D} = \frac{1}{(0.01)(2)} = 50\text{ k}\Omega$ $A_{v(\text{new})} = -g_m (R_D || r_o) = -2(3.6 || 50) = -6.7$

Problem 3 : An NMOS amplifier is to be designed to provide a 0.5V peak output signal across a 50k Ω load that can be used as a drain resistor. If a gain of at least 5 V/V is needed, what g_m is required? Using a dc supply of 3 V, what values of I_D and V_{OV} would you choose if $V_t = 0.8\text{ V}$, find V_{GS} ? What (W/L) ratio is required if $\mu_n C_{ox} = 100\text{ }\mu\text{A/V}^2$?

Solution: Given $A_v = 5$; $R_D = 50\text{ k}\Omega$ $\rightarrow g_m = \frac{A_v}{R_D} = \frac{5}{50\text{ k}} = 0.1\text{ mS}$

For 0.5V output, the required input signal is given by $v_{gs} = \frac{v_o}{A_v} = \frac{0.5}{5} = 0.1 \text{ V}$

Hence $V_{DS} - 0.5 \geq V_{GS} + v_{gs} - V_t$

$V_{DS} \geq V_{GS} + 0.1 - 0.8 + 0.5$ or $V_{DS} \geq V_{GS} - 0.2$ ----- (a)

Considering the second aspect $V_{DS} + 0.5 \leq V_{DD}$ or $V_{DS} + 0.5 \leq 3$

$V_{DS} \leq 2.5$ ----- (b)

The circuit has to be designed for lowest possible V_{DS} so that the device operates in saturation region.

Hence choose equation (a).

Using $V_{DS} = V_{DD} - I_D R_D$ ----- (c)

Equating to equation (a) and (c) $\rightarrow V_{DD} - I_D R_D = V_{GS} - 0.2$ or $I_D = \frac{V_{DD} + 0.2 - V_{GS}}{R_D} = \frac{3.2 - V_{GS}}{50k}$

Also $g_m = 0.1 = \left[\frac{2I_D}{V_{GS} - V_t} \right] = \left[\frac{2 \left[\frac{3.2 - V_{GS}}{50} \right]}{V_{GS} - 0.8} \right] \rightarrow 2.5V_{GS} - 2 = 3.2 - V_{GS}$ or $V_{GS} = \frac{5.2}{3.5} = 1.49 \text{ V}$

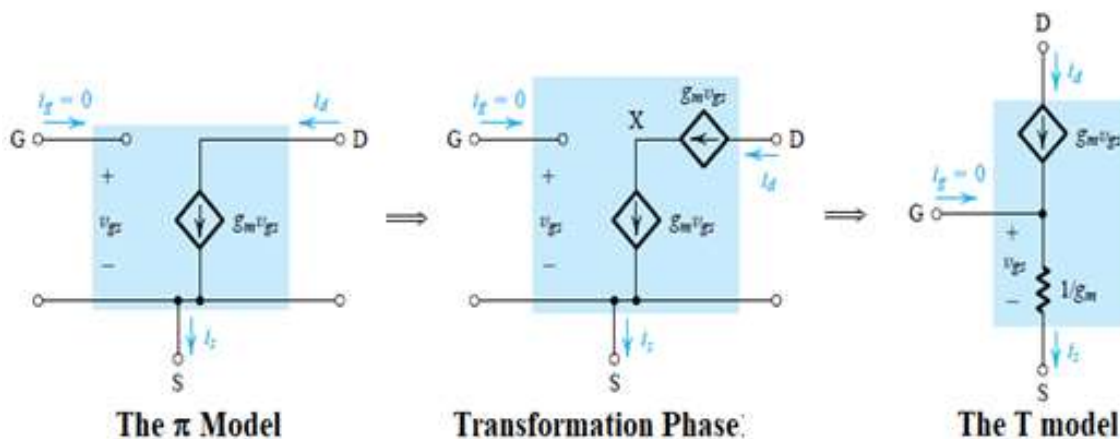
$I_D = \frac{3.2 - V_{GS}}{50k} = \frac{3.2 - 1.49}{50k} = 0.034 \text{ mA}$ and $V_{DS} = V_{GS} - 0.2 = 1.49 - 0.2 = 1.29 \text{ V}$

Overdrive voltage $V_{OV} = V_{GS} - V_t = 1.49 - 0.8 = 0.69 \text{ V}$

$\left(\frac{W}{L} \right) = \frac{g_m^2}{2k_n I_D} = \frac{1 \times 10^{-6}}{2 \times 100 \times 10^{-6} (0.034 \times 10^{-3})} = 1.47$

The T Model

The T model is preferred when there exists a resistance in the source circuit. The illustrations below shows how a MOSFET π - model into an equivalent T – model.



Review Questions

1. Draw the conceptual MOSFET amplifier circuit and derive the expression for (i) small signal drain current i_D (ii) Transconductance g_m and (iii) voltage gain A_v
2. With the small signal equivalent model of MOSFET, derive an expression for voltage gain and transconductance.
3. Draw and explain the small signal model of the MOSFET amplifier, assuming $\lambda \neq 0$.
4. Mention the design parameters for a MOSFET amplifier and how transconductance can be computed using these parameters.

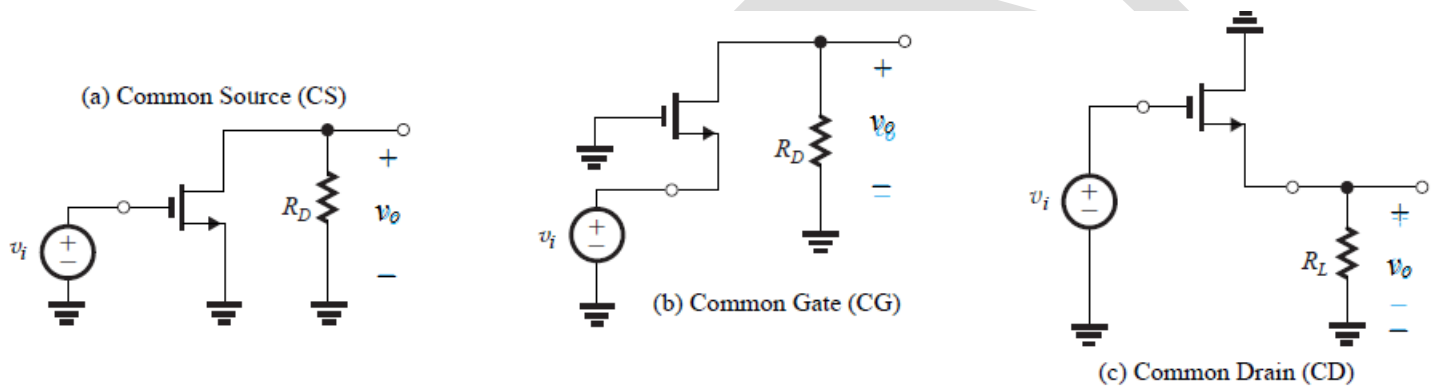
Module 2(c) - MOSFET Amplifier Configurations

Syllabus - MOSFET Amplifier configuration: Basic configurations, characterizing amplifiers, CS amplifier with and without source resistance. The Common Gate Amplifier, Source follower.

2.3 - Basic MOSFET Amplifier Configurations

By biasing the MOSFET at a suitable point in its saturation region of operation and by keeping the input signal small, almost-linear amplification can be obtained. There are three basic configurations for connecting the MOSFET as an amplifier. Each of these configurations is obtained by connecting one of the three MOSFET terminals to ground, thus creating a two-port network with the grounded terminal being common to the input and output ports. The three configurations are

- (i) Common Source (CS) amplifier (ii) Common Gate (CG) amplifier and
(iii) Common Drain (CD) amplifier or Source follower.



2.3(1) - Characterizing Amplifiers

Figure shows an amplifier fed with a signal source having an open-circuit voltage and an internal resistance R_{sig} . The amplifier is shown with a load resistance R_L connected to the output terminal. Here, R_L can be an actual load resistance or the input resistance of a succeeding amplifier stage in a cascade amplifier.

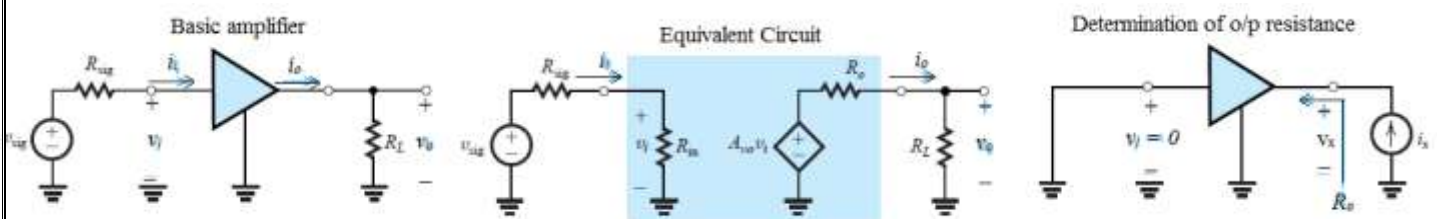
The input resistance R_{in} represents the loading effect of the amplifier input on the signal source.

(i) Input Resistance $R_{in} = \frac{v_i}{i_i}$

The amplifier input $v_i = \left[\frac{R_{in}}{R_{sig} + R_{in}} \right] v_{sig}$ and $v_{sig} = \left[\frac{R_{sig} + R_{in}}{R_{in}} \right] v_i$

(ii) Open circuit voltage gain $A_{vo} = \left[\frac{v_o}{v_i} \right]_{R_L = \infty}$

(iii) The output resistance R_o is the resistance seen looking back into the amplifier output terminal with $v_i = 0$



(iii) Output Resistance $R_o = \frac{V_x}{i_x}$

From the Thevenin's equivalent circuit,

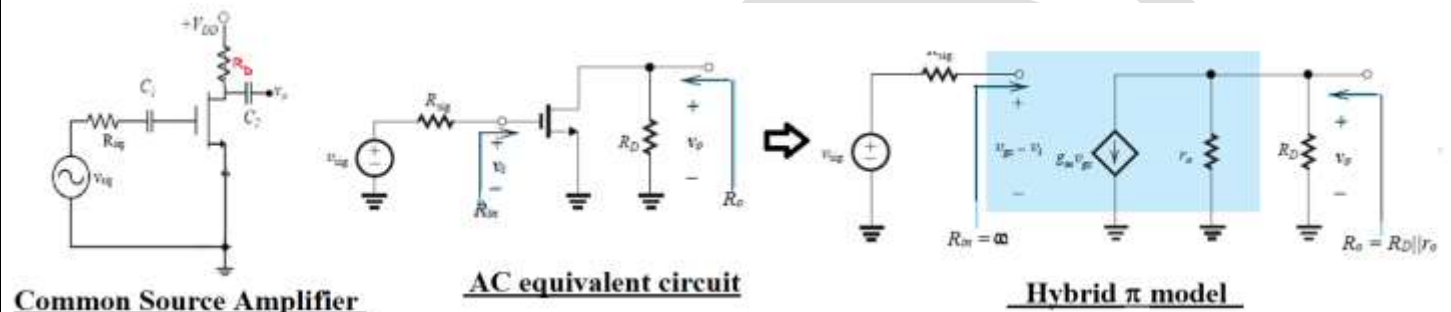
(iv) Output voltage $V_o = \left[\frac{R_L}{R_L + R_o} \right] A_{vo} v_i$

(v) Hence the voltage gain of the proper amplifier $A_v = \frac{v_o}{v_i} = \left[\frac{R_L}{R_L + R_o} \right] A_{vo}$

(vi) The overall gain $G_v = \frac{v_o}{v_{sig}} = \left[\frac{R_{in}}{R_{sig} + R_{in}} \right] \frac{1}{v_i} \left[\frac{R_L}{R_L + R_o} \right] A_{vo} v_i = \left[\frac{R_{in}}{R_{sig} + R_{sig}} \right] \left[\frac{R_L}{R_L + R_o} \right] A_{vo}$

2.3(2) - The Common-Source (CS) Amplifier without source resistance(source resistance bypassed)

It is the most widely used MOS amplifier configuration and can be used in stages to achieve the desired voltage gain. The source terminal is made common between input and output as shown in the figure below. It is assumed that the drain resistance R_D is a part of the amplifier and the external load resistor R_L appears in parallel with R_D .



Consider the Hybrid model equivalent of the amplifier;

(i) Input Resistance R_{in} : Due to open circuit at the input side (gate and source are isolated) $R_{in} = \infty$

(ii) The output voltage $v_o = -(g_m v_{gs})(r_o || R_D) = -(g_m v_i)(r_o || R_D)$ (since $v_{gs} = v_i$)

(iii) Open circuit Voltage gain $A_{vo} = \left[\frac{v_o}{v_i} \right]_{R_L = \infty} = -g_m (r_o || R_D) \cong -g_m R_D$ (since $R_D \ll r_o$)

(iv) The output resistance R_o is obtained by setting input $v_i = 0$; hence $v_{gs} = 0$ and $g_m v_{gs} = 0$; hence the current source acts as a open circuit.

$R_o = (r_o || R_D) \cong R_D$

From the above analysis of the CS amplifier, the following observations are made:

1. The input resistance R_{in} is ideally infinite. 2. The output resistance is moderate to high. 3. The open-circuit voltage gain A_{vo} can be high. 4. The bandwidth of the CS amplifier is severely limited.

Overall Voltage Gain G_v : The overall voltage gain is determined by considering the load resistance R_L

Voltage gain $A_v = \frac{v_o}{v_i} = -g_m (r_o || R_D || R_L)$ since $v_{gs} = v_i$

Over all voltage gain $G_v = A_v = -g_m (r_o || R_D || R_L)$

Problem 1: A CS amplifier utilizes a MOSFET biased at $I_D = 0.25$ mA with $V_{ov} = 0.25$ V and $R_D = 20$ k Ω . The device has $V_A = 50$ V. The amplifier is fed with a source having $R_{sig} = 100$ k Ω , and a 20 k Ω load is connected to the output. Find R_{in} , A_{vo} , R_o , A_v , and G_v . If to maintain reasonable linearity, the peak of the input sine-wave signal is limited to 10% of $(2V_{ov})$, what is the peak of the sine wave voltage at the output?

Solution: Since it is CS amplifier; $R_{in} = \infty$

$$r_o = \left| \frac{V_A}{I_D} \right| = \frac{50}{0.25 \times 10^{-3}} = 200 \text{ k}\Omega \quad \text{and} \quad g_m = \frac{2I_D}{V_{ov}} = \frac{2 \times 0.25 \times 10^{-3}}{0.25} = 2 \text{ mA/V}$$

$$A_{vo} = -g_m (r_o || R_D) = (2 \times 10^{-3})(200k || 20k) = -36.36$$

$$R_o = (r_o || R_D) \cong 18.18 \text{ k}\Omega$$

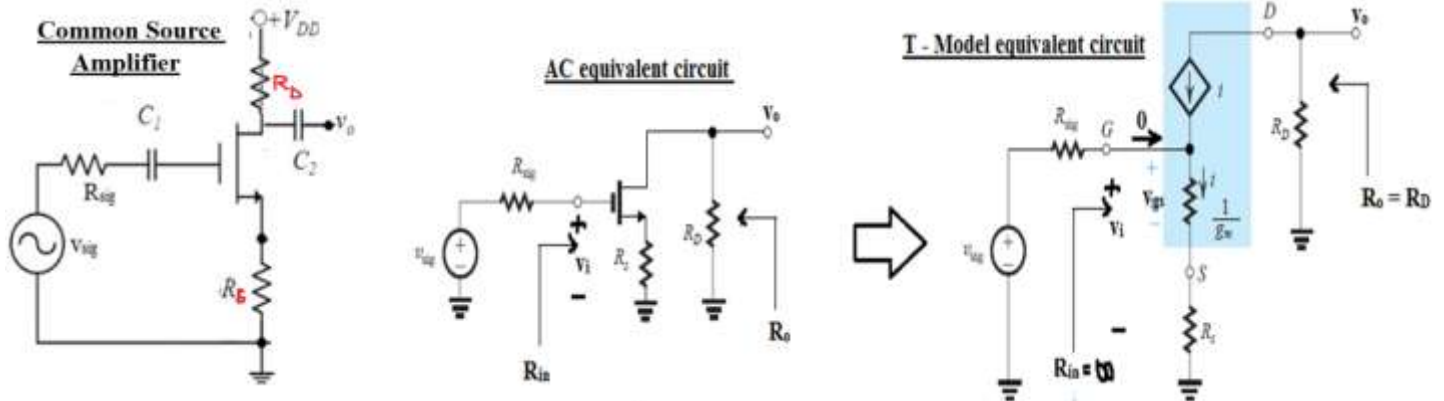
$$\text{Over all voltage gain } G_v = A_v = -g_m (r_o || R_D || R_L) = (2 \times 10^{-3})(200k || 20k || 20k) = -19.05$$

$$\text{For } v_{gs} = 10\%(2V_{ov}) = 0.1(2)(0.25) = 0.05 \text{ V}$$

$$v_o = |A_v v_{gs}| = 19.05(0.05) = 0.9525 \text{ V}$$

2.3(3)-The Common-Source (CS) Amplifier with a Source Resistance R_S (Source resistance unbypassed)

Figure shows the Common Source (CS) amplifier with source resistance R_S . The source resistance R_S stabilizes the Q point. In general, whenever a resistance is connected in the source lead, the T model is preferred. The source resistance then simply appears in series with the resistance $(1/g_m)$ and can be added to it.



(i) **Input Resistance R_{in} :** Due to open circuit at the input side (gate and source are isolated); $R_{in} = \infty$

(ii) **The output voltage:** Due to open circuit at the input side (gate and source are isolated) $v_i = v_{sig}$

From the equivalent circuit; applying voltage divider rule, the voltage v_{gs} can be expressed in terms of input voltage v_i and is given as

$$v_{gs} = \left[\frac{v_i \left(\frac{1}{g_m} \right)}{\left(\frac{1}{g_m} \right) + R_S} \right] = \left[\frac{v_i}{1 + g_m R_S} \right] \quad \text{and} \quad v_o = -i R_D = - \left[\frac{v_i}{\left(\frac{1}{g_m} \right) + R_S} \right] R_D = - \left[\frac{g_m v_i}{1 + g_m R_S} \right] R_D$$

(iii) **Small signal voltage gain** $A_v = \frac{v_o}{v_i} = \frac{-g_m R_D}{(1 + R_S g_m)}$

(iv) The output resistance R_o is obtained by setting input $v_i = 0$; hence $R_o = R_D$

(v) If the load resistance R_L is connected; then

$$\text{The overall gain } A_v = - \left[\frac{g_m (R_D || R_L)}{1 + R_S g_m} \right]$$

Benefits of using source resistance R_S :

(i) R_S can be used to control the magnitude of the signal v_{gs} and thereby ensure that v_{gs} does not become too large and cause unacceptably high nonlinear distortion. (ii) R_S provide the negative feedback, ensuring increase in bandwidth of the amplifier. (iii) R_S stabilizes the operating point.

Problem 2: A common source amplifier without source resistance having an input signal of 50 mV(peak) and an output signal of 1V (peak). Now if the input signal is changed to 0.2V (peak), what value of source resistance to be added to keep the output signal same as 1 V(peak). Assume $g_m = 2 \text{ mA/V}$

Solution: Gain without source resistance $A_V = \frac{V_o}{V_{in}} = \frac{1}{50 \times 10^{-3}} = 20 = -g_m(R_D || R_L) \text{ --- (1)}$

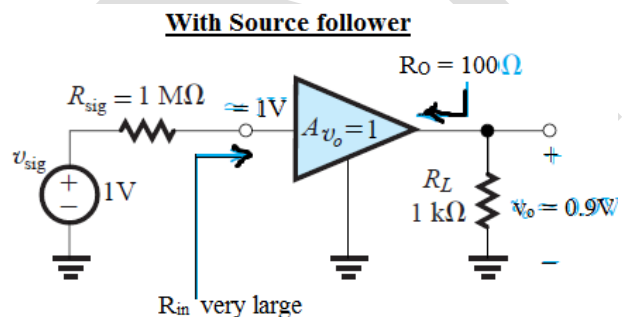
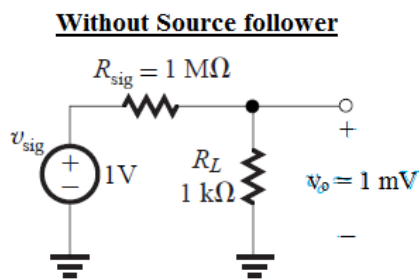
Gain with source resistance $A_V = \frac{V_o}{V_{in}} = \frac{1}{0.2} = 5 = -\left[\frac{g_m(R_D || R_L)}{1 + R_s g_m}\right] \text{ --- (2)}$

Dividing Eqn(1) by eqn(2) results $(1 + R_s g_m) = (20/5) = 4$ or $R_s g_m = 3$

Hence $R_s = \frac{3}{g_m} = \frac{3}{2 \times 10^{-3}} = 1.5 \text{ k}\Omega$

2.3(4) Need for Voltage Buffers or Source follower

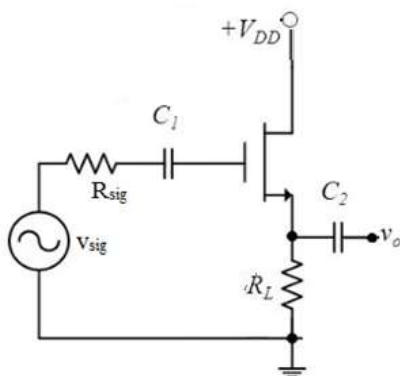
A signal source delivering a signal of reasonable strength (1 V) directly to a load of 1 k Ω shown in the figure. Severe attenuation of the signal takes place at the output and the signal appearing across the load will be only $\left(\frac{1}{1000+1}\right)$ of the input signal or about 1 mV.



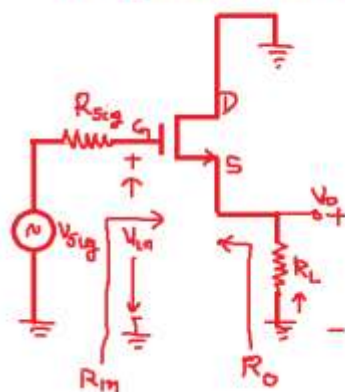
If an unity gain amplifier with high input resistance and low output resistance is interposed between the source and the load, almost all of v_{sig} (i.e., 1 V) will appear at the input of the amplifier proper. Since the amplifier has low output resistance (100 Ω), 90% of this signal (0.9 V) will appear at the output. The source follower is basically a unity gain amplifier.

2.3(5) - The Common-Drain Amplifier or Source Follower

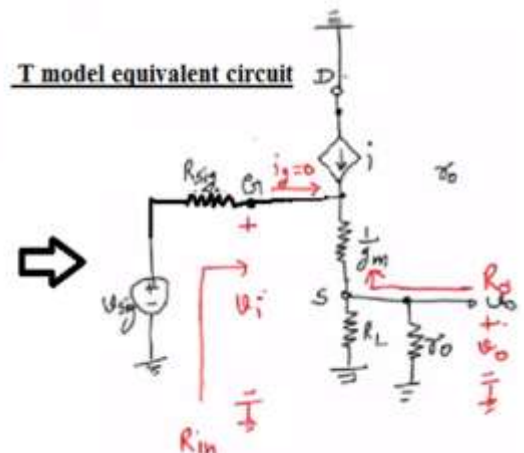
The common-drain amplifier is an important circuit that finds application in the design of both small-signal amplifiers as well as amplifiers that are required to handle large signals and deliver substantial amounts of signal power to a load. The common drain amplifier is more commonly known as the source follower. Figure shows the circuit diagram of an NMOS source follower; the output is taken across the source with respect to the ground and the drain is connected directly to V_{DD} .



AC equivalent circuit



T model equivalent circuit



Analysis using T model : Figure shows the source follower ac equivalent circuit and its corresponding T model equivalent circuit.

Input Resistance R_{in} Since $i_g = 0$; $v_{in} = v_{sig}$ and Input resistance $R_{in} = \infty$

Voltage gain A_v : Since $i_g = 0$; $v_{in} = v_{sig}$ Hence Voltage gain A_v is same as overall voltage gain G_v

From the equivalent circuit; applying voltage divider rule, the output voltage v_o can be expressed in terms of input voltage v_i and is given as

$$v_o = \left[\frac{v_{in}(r_o || R_L)}{\left(\frac{1}{g_m}\right) + (r_o || R_L)} \right] = \left[\frac{v_{in}(r_o || R_L)}{\left(\frac{1}{g_m}\right) + (r_o || R_L)} \right] = \left[\frac{v_{sig} R_L}{\left(\frac{1}{g_m}\right) + R_L} \right]_{r_o \gg R_L}$$

$$\text{Voltage gain } A_v = \frac{v_o}{v_{in}} = \left[\frac{R_L}{\left(\frac{1}{g_m}\right) + R_L} \right] = \left[\frac{g_m R_L}{1 + g_m R_L} \right] = \left[\frac{g_m}{g_m + \left(\frac{1}{R_L}\right)} \right] = G_v$$

$$\text{The open circuit Voltage gain } A_{v_o} = \left[\frac{v_o}{v_{sig}} \right]_{R_L = \infty} = \left[\frac{g_m}{g_m + \left(\frac{1}{R_L}\right)} \right] = 1 \quad \text{--- -- (hence the name source follower)}$$

Output Resistance R_o : To determine R_o ; Set the input $v_{in} = 0$; hence only resistance $(1/g_m)$ will be present in the circuit. Output resistance $R_o = \left(\frac{1}{g_m}\right)$

In conclusion, the source follower features a very high input resistance (ideally, infinite), a relatively low output resistance, and an open-circuit voltage gain that is near unity (ideally, unity). Thus the source follower is ideally suited for implementing the unity-gain voltage buffer. The source follower is also used as the output (i.e., last) stage in a multistage amplifier, where its function is to equip the overall amplifier with a low output resistance, thus enabling it to supply relatively large load currents without loss of gain (i.e., with little reduction of output signal level).

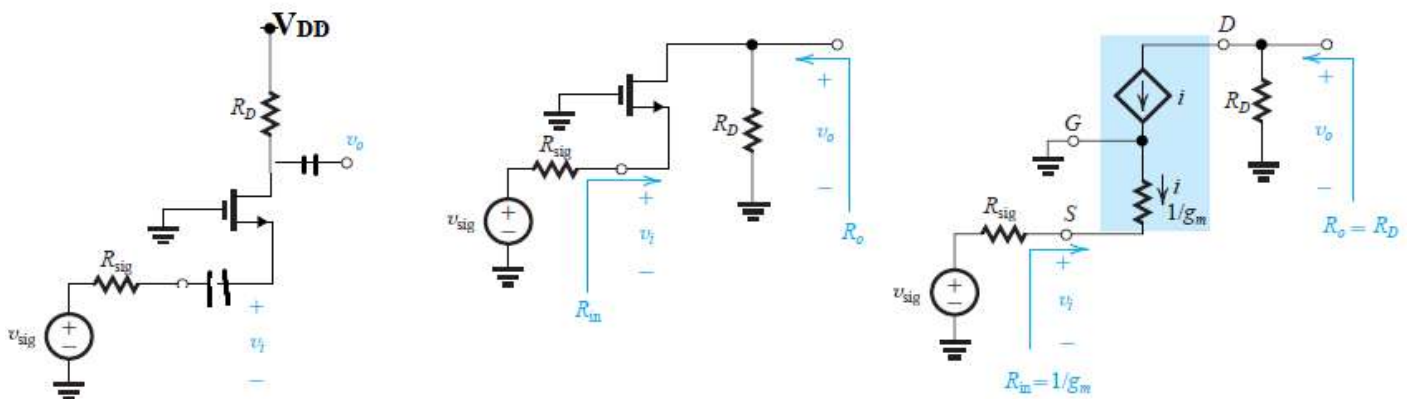
Common Gate (CG) Amplifier

Figure shows a Common Gate amplifier, its ac equivalent circuit and its corresponding T – model where in the gate is grounded. The input is applied between Source and gate and out is taken across the drain resistance R_D .

The Common-Gate (CG) Amplifier

Ac Equivalent Circuit

T Model



(1) Input Resistance R_{in} - The input resistance is between Source and ground. Gate is grounded,

$$\text{By inspection } R_{in} = \frac{1}{g_m} \text{ ohms}$$

The value of R_{in} is few hundreds of ohms, hence the input resistance of CG amplifier is less.

(2). **Voltage Gain** – The source current $i = -\left(\frac{v_i}{\frac{1}{g_m}}\right) = -g_m v_i$

and $v_o = -i [R_D || r_o] = -(-g_m v_i) R_D = v_i g_m R_D$ ----- ($r_o \gg R_D$)

$$\text{and } A_{vo} = \frac{v_o}{v_i} = g_m R_D$$

The input voltage to amplifier v_i can be expressed in terms of v_{sig} using voltage divider rule as

$$v_i = i \left(\frac{1}{g_m}\right) = \left(\frac{v_{sig}}{\frac{1}{g_m} + R_{sig}}\right) \left(\frac{1}{g_m}\right) = \left[\frac{v_{sig}}{1 + g_m R_{sig}}\right] = \left[\frac{v_{sig} R_{in}}{R_{in} + R_{sig}}\right]$$

Since $V_{sig} = v_i$; The overall voltage gain and voltage gain are same

The **Overall voltage gain G_v** with load resistance R_L is connected can be expressed as

$$G_v = A_v = \frac{v_o}{v_{sig}} = \frac{v_i g_m (R_D || R_L)}{v_{sig}} = \left(\frac{v_{sig} R_{in}}{R_{in} + R_{sig}}\right) \left(\frac{g_m (R_D || R_L)}{v_{sig}}\right) = \left(\frac{\left(\frac{1}{g_m}\right)}{\left(\frac{1}{g_m}\right) + R_{sig}}\right) (g_m (R_D || R_L))$$

$$G_v = \frac{v_o}{v_{sig}} = \frac{(R_D || R_L)}{\left(\frac{1}{g_m}\right) + R_{sig}} = \frac{g_m (R_D || R_L)}{1 + g_m R_{sig}}$$

The output resistance R_o

By inspection $R_o = R_D$

Problem 3- A CG amplifier is required to match a signal source with $R_{sig} = 100\Omega$. At what current should the MOSFET be biased if it is operated at an overdrive voltage of 0.2 V? If the total resistance in the drain circuit is 2 k, what overall voltage gain is realized?

Solution Given that the Amplifier is matched to R_{sig} hence $R_{in} = (1/g_m) = R_{sig} = 100\Omega$ or $g_m = 10 \text{ mS}$

Using the relation

$$I_D = \frac{g_m V_{ov}}{2} = \frac{0.2 \times 10 \times 10^{-3}}{2} = 1 \text{ mA} \quad \text{and} \quad G_v = \frac{g_m R_D}{1 + g_m R_{sig}} = \frac{10 \times 10^{-3} (2 \times 10^3)}{1 + (10 \times 10^{-3})(100)} = 10$$

Review Questions

1. Define the terms input resistance, output resistance, voltage gain and overall gain of an amplifier.
2. With a neat circuit diagram and ac equivalent circuit, derive the expression for R_{in} , A_v , and R_o for a CS amplifier with bypassed source resistance.
3. With a neat circuit diagram and ac equivalent circuit derive the expressions for R_{in} , A_{vo} , A_v and R_o for common source amplifier with an un-bypassed source resistance.
4. Analyze the circuit of common drain amplifier and derive the expressions for no-load voltage gain, overall voltage gain, input resistance and output resistance.
5. With a neat circuit diagram and ac equivalent circuit, derive the expressions for R_{in} , A_{vo} , A_v and R_o for a Source follower.
6. With a neat circuit diagram and ac equivalent circuit, derive the expressions for R_{in} , A_{vo} , A_v and R_o for a Common Gate Amplifier.